

Laboratory 2.1: Oncogene and Tumor Suppressor Analysis

Good morning, everyone. Welcome to our second laboratory session in molecular cancer biology. Today marks a significant step forward in our practical exploration of cancer mechanisms. While last week we observed cancer cells and characterized their basic properties, today we're going to peer directly into their molecular machinery to examine the genetic and protein alterations that drive their malignant behavior.

Over the next four hours, you'll be working with the same three cell lines from last week—HeLa, MCF-7, and HCT116—but now we're going to analyze them at the molecular level. We'll examine KRAS mutations using PCR and DNA sequencing, and we'll assess the protein levels and cellular localization of three crucial tumor suppressors: p53, APC, and RB.

This laboratory will give you hands-on experience with some of the most important techniques in cancer research while reinforcing the concepts we've discussed in lectures about oncogenes and tumor suppressors.

Laboratory Overview and Molecular Strategy

Today's work combines several complementary approaches to molecular analysis. We're using DNA-based techniques to detect oncogene mutations and protein-based techniques to assess tumor suppressor status. This dual approach reflects how modern cancer research integrates multiple levels of molecular analysis to understand cancer biology.

The choice of KRAS as our target oncogene and p53, APC, and RB as our tumor suppressors isn't arbitrary. These represent some of the most clinically important cancer genes, and each of your cell lines has known alterations in these pathways that we can detect and analyze.

Understanding the molecular profiles of these cell lines will help you appreciate how different genetic alterations contribute to the distinct biological behaviors you observed last week. More importantly, you'll gain experience with the techniques that are used daily in cancer research laboratories and increasingly in clinical diagnostics.

Part 1: PCR Amplification of KRAS Hotspot Mutations

We'll begin with PCR amplification of KRAS hotspot regions, focusing on codons 12, 13, and 61. These three codons account for the vast majority of oncogenic KRAS mutations found in human cancers.

Understanding KRAS Hotspot Mutations

Before we start the molecular work, let's review why these particular codons are so important. KRAS functions as a molecular switch that cycles between active GTP-bound and inactive GDP-bound states. The protein has intrinsic GTPase activity that normally allows it to hydrolyze GTP to GDP, turning itself off.

Mutations at codons 12, 13, and 61 impair this GTPase activity without affecting GTP binding. The result is a KRAS protein that gets stuck in the active, GTP-bound state, continuously sending proliferative signals to the cell.

Codon 12 mutations are the most common, found in about 80% of pancreatic cancers and 40% of colorectal cancers. Codon 13 mutations are less frequent but still clinically significant. Codon 61 mutations are rarer but equally important when they occur.

DNA Extraction Quality Assessment

Before we begin PCR, let's assess the quality of the DNA you extracted last week. Good PCR requires high-quality template DNA that's free from contaminating proteins and chemicals that might inhibit the polymerase.

Take a small aliquot of each DNA sample and assess its concentration and purity using the spectrophotometer. We're looking for A260/A280 ratios between 1.8 and 2.0, which indicate relatively pure DNA. Lower ratios suggest protein contamination, while higher ratios might indicate RNA contamination or alkaline conditions.

Also run a small sample on an agarose gel to assess DNA integrity. You should see a high molecular weight band with minimal smearing. Extensive smearing indicates DNA degradation, which can reduce PCR efficiency.

Primer Design Principles

The primers we're using today were designed to amplify short regions around each KRAS hotspot. Let me explain the design principles so you understand how these primers work.

For codon 12 and 13, which are located in exon 2, we use primers that flank this region with adequate margin for sequencing. The forward primer is located in the intron upstream of exon 2, while the reverse primer is in the downstream intron. This ensures we capture the entire exon and can detect any mutations within it.

For codon 61, located in exon 3, we use a similar flanking strategy. The amplicons are designed to be 200-300 base pairs long—large enough to provide good sequencing context but small enough for efficient PCR amplification.

The primers are also designed to have similar melting temperatures so they can be used under the same PCR conditions. They avoid regions of secondary structure or repetitive sequences that might cause non-specific amplification.

PCR Setup and Thermocycling

Now let's set up our PCR reactions. We'll be using a high-fidelity DNA polymerase that has 3' to 5' exonuclease activity for proofreading. This reduces the error rate and ensures that any mutations we detect are real rather than PCR artifacts.

For each DNA sample, set up three separate PCR reactions—one for each hotspot region. Use the same master mix for consistency, but add the appropriate primer pair for each reaction.

The master mix contains:

- DNA polymerase with proofreading activity
- dNTPs at appropriate concentrations
- Buffer with magnesium chloride optimized for this polymerase
- Loading dye for easy gel visualization

Add 50-100 nanograms of template DNA to each reaction. Too little DNA will reduce amplification efficiency, while too much can lead to non-specific amplification.

Thermocycling Parameters

Our PCR protocol uses a three-step cycling approach:

Initial denaturation at 95 degrees Celsius for 3 minutes ensures complete denaturation of the template DNA and activates the hot-start polymerase.

Cycling phase repeats 35 times:

- Denaturation at 95 degrees for 30 seconds
- Annealing at 58 degrees for 30 seconds
- Extension at 72 degrees for 45 seconds

The annealing temperature of 58 degrees was empirically determined to provide specific amplification of our target sequences while minimizing non-specific products.

Final extension at 72 degrees for 5 minutes ensures complete synthesis of all amplicons.

Gel Electrophoresis Analysis

While the PCR is running, let's prepare our agarose gel for analyzing the products. We'll use a 2% agarose gel, which provides good resolution for our expected product sizes.

Dissolve the agarose in TBE buffer by microwaving until completely melted. Allow it to cool slightly before adding ethidium bromide or a safer alternative like SYBR Safe for DNA visualization.

Pour the gel and insert the comb, making sure there are no air bubbles. The gel should be about 5 millimeters thick for optimal resolution.

When the PCR is complete, load the entire reaction volume onto the gel along with a DNA ladder that spans the expected product sizes. Run the gel at 100-120 volts until the dye front has migrated about two-thirds of the way down the gel.

Expected Results and Troubleshooting

You should see single, sharp bands at the expected sizes for each primer pair:

- Codon 12/13 amplicon: approximately 250 base pairs
- Codon 61 amplicon: approximately 200 base pairs

Multiple bands suggest non-specific amplification, which could indicate suboptimal primer concentrations or annealing temperatures. Weak or absent bands suggest insufficient template DNA or PCR inhibition.

If you see primer dimers—small bands around 50-100 base pairs—this indicates excess primer concentration. These shouldn't interfere with our analysis as long as the main products are clearly visible.

Part 2: Sanger Sequencing Analysis

Once we've confirmed successful PCR amplification, we'll prepare our products for Sanger sequencing analysis.

PCR Product Purification

Before sequencing, we need to remove excess primers and dNTPs that would interfere with the sequencing reaction. We'll use a combination of enzymatic cleanup and column purification.

First, treat each PCR product with exonuclease I and shrimp alkaline phosphatase. Exonuclease I degrades single-stranded DNA, including excess primers, while shrimp alkaline phosphatase removes phosphate groups from dNTPs, making them unavailable for the sequencing reaction.

Incubate this mixture at 37 degrees for 15 minutes, then heat-inactivate the enzymes at 80 degrees for 15 minutes.

For even cleaner results, you can also purify the products using spin columns. These columns retain DNA while allowing smaller molecules like primers and dNTPs to pass through.

Sequencing Reaction Setup

We'll use the chain termination method for DNA sequencing, employing fluorescently labeled dideoxynucleotides. The sequencing reaction contains:

- Purified PCR product as template
- Sequencing primer (either forward or reverse)
- DNA polymerase optimized for sequencing
- dNTPs for normal chain elongation
- Fluorescent ddNTPs for chain termination

The ratio of dNTPs to ddNTPs is carefully calibrated to produce a range of termination products that cover the entire region of interest.

Set up both forward and reverse sequencing reactions for each PCR product. Sequencing from both directions provides confirmation of any mutations detected and helps identify sequencing artifacts.

Capillary Electrophoresis and Data Collection

The sequencing products will be analyzed by capillary electrophoresis, which separates the fluorescently labeled DNA fragments by size with single-nucleotide resolution.

As the DNA fragments migrate through the capillary, they pass a laser that excites the fluorescent labels. Each of the four ddNTPs has a different fluorescent label, allowing the instrument to identify which nucleotide terminated each fragment.

The result is a chromatogram showing fluorescent peaks corresponding to each nucleotide position. The height and quality of these peaks indicate the confidence of the base call.

Part 3: Mutation Calling and Analysis

Now comes the critical step of analyzing the sequencing data to identify mutations.

Chromatogram Interpretation

Load your sequencing files into appropriate software for analysis. You'll see chromatograms with four-color traces representing the four nucleotides.

High-quality sequence shows:

- Sharp, well-separated peaks
- Consistent peak heights within each trace
- Minimal background fluorescence
- Clear baseline separation between peaks

Poor-quality regions show:

- Broad, overlapping peaks
- Inconsistent peak heights
- High background fluorescence
- Noisy baselines

Identifying Mutations

To identify mutations, compare your sequences to the reference KRAS sequence. Pay particular attention to the codon positions we're analyzing:

Codon 12 (GGT in wild-type): Look for changes to GAT (G12D), GTT (G12V), TGT (G12C), or other variants.

Codon 13 (GGC in wild-type): Look for changes to GAC (G13D) or other variants.

Codon 61 (CAA in wild-type): Look for changes to CGA (Q61R), CAT (Q61H), or other variants.

Mutations appear as different-colored peaks at the expected positions. Heterozygous mutations show overlapping peaks of two different colors at the same position.

Confirming Mutations

Any potential mutations should be confirmed by:

- Sequencing from the opposite direction
- Repeating the PCR and sequencing from fresh DNA
- Comparing results across multiple cell lines

Remember that cell lines are typically homozygous for mutations due to loss of heterozygosity, so you should see clean substitution peaks rather than overlapping heterozygous patterns.

Expected Results for Cell Lines

Based on published data, you should expect to find:

- **HeLa cells:** Wild-type KRAS at all three codons
- **HCT116 cells:** KRAS G13D mutation (codon 13 GGC→GAC)
- **MCF-7 cells:** Wild-type KRAS at all three codons

If your results don't match these expectations, consider technical issues like PCR contamination, sample mix-ups, or sequencing problems.

Part 4: Western Blotting for Tumor Suppressor Proteins

Now we'll shift from DNA analysis to protein analysis, examining the levels of p53, APC, and RB proteins in our cell lines.

Protein Extraction and Quantification

First, we need to extract total cellular proteins from each cell line. We'll use a standard RIPA buffer that contains detergents to solubilize membrane proteins and salts to extract nuclear proteins.

Harvest cells by trypsinization as you did last week, but this time wash the cells twice with cold PBS to remove any serum proteins that might interfere with our analysis.

Resuspend the cell pellet in RIPA buffer containing protease inhibitors. The protease inhibitors are crucial because they prevent protein degradation during extraction and storage.

Lyse the cells by incubating on ice for 20 minutes with occasional vortexing. Complete lysis is important for quantitative protein extraction.

Clear the lysate by centrifugation at high speed for 10 minutes at 4 degrees. The supernatant contains your soluble proteins, while the pellet contains debris and insoluble material.

Protein Quantification

Accurate protein quantification is essential for western blotting because we need to load equal amounts of protein in each lane for meaningful comparisons.

We'll use the Bradford assay, which is based on the binding of Coomassie Brilliant Blue dye to proteins. The dye undergoes a color change when bound to protein, shifting from brown to blue.

Prepare a standard curve using known concentrations of bovine serum albumin. This allows you to determine the protein concentration in your samples by comparing their absorbance to the standard curve.

Dilute your protein samples appropriately—typically 1:10 or 1:20—to ensure they fall within the linear range of the assay.

SDS-PAGE Setup

Sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) separates proteins by molecular weight under denaturing conditions.

We'll use pre-cast gradient gels that provide good resolution across a wide range of molecular weights. The gradient allows smaller proteins to migrate further while retaining larger proteins in the upper portion of the gel.

Load equal amounts of protein (typically 20-40 micrograms) in each lane. Include a molecular weight marker to identify protein sizes and a positive control if available.

Add loading buffer containing SDS, beta-mercaptoethanol, and bromophenol blue. SDS denatures proteins and gives them uniform negative charge, beta-mercaptoethanol reduces disulfide bonds, and bromophenol blue tracks the electrophoresis progress.

Electrophoresis and Transfer

Run the gel at constant voltage, starting at 80-100 volts until the samples enter the resolving gel, then increasing to 120-150 volts for faster separation.

Monitor the progress using the dye front. Stop the electrophoresis when the dye front reaches the bottom of the gel but before proteins run off.

Transfer proteins to a nitrocellulose or PVDF membrane using wet transfer or semi-dry transfer methods. The transfer efficiency can be checked by staining the membrane with Ponceau S, which reversibly stains all proteins.

Immunoblotting Procedure

Block the membrane with 5% non-fat dry milk or BSA in TBS-T (Tris-buffered saline with Tween-20) for at least one hour. This prevents non-specific antibody binding.

Incubate with primary antibodies overnight at 4 degrees or for 2 hours at room temperature. We'll use antibodies against:

- **p53**: Recognizes both wild-type and mutant forms
- **APC**: Recognizes the N-terminal region that's retained in most truncating mutations
- **RB**: Recognizes the hypophosphorylated form

After primary antibody incubation, wash extensively with TBS-T to remove unbound antibody.

Incubate with horseradish peroxidase-conjugated secondary antibodies for 1 hour at room temperature. These antibodies recognize the primary antibodies and provide the enzymatic activity for detection.

Detection and Analysis

Detect bound antibodies using enhanced chemiluminescence (ECL). The horseradish peroxidase catalyzes the oxidation of luminol in the presence of hydrogen peroxide, producing light that can be captured on film or by digital imaging.

Expose the membrane for various times to optimize signal intensity. Overexposure leads to saturated bands, while underexposure may miss weak signals.

Include a loading control antibody (like actin or tubulin) to normalize for loading differences between lanes.

Part 5: Immunofluorescence Microscopy

Finally, we'll examine the subcellular localization of our tumor suppressor proteins using immunofluorescence microscopy.

Cell Preparation and Fixation

Plate cells on coverslips or chamber slides 24-48 hours before the experiment to allow them to attach and spread properly.

Fix cells with 4% paraformaldehyde for 15 minutes at room temperature. Paraformaldehyde crosslinks proteins while preserving cellular morphology and protein localization.

Permeabilize cells with 0.1% Triton X-100 for 10 minutes to allow antibodies access to intracellular proteins.

Immunostaining Protocol

Block cells with 5% normal serum (from the species in which the secondary antibody was raised) for 1 hour to prevent non-specific binding.

Incubate with primary antibodies diluted in blocking buffer. Use the same antibodies as for western blotting, but typically at lower concentrations since we're looking at localization rather than quantification.

Incubate overnight at 4 degrees or for 2 hours at room temperature in a humidified chamber to prevent evaporation.

Wash extensively with PBS to remove unbound primary antibody.

Incubate with fluorescently labeled secondary antibodies for 1 hour at room temperature in the dark. Choose secondary antibodies with non-overlapping emission spectra if doing multiple stains.

Nuclear Counterstaining and Mounting

Counterstain nuclei with DAPI (4',6-diamidino-2-phenylindole) or Hoechst dye to visualize nuclear boundaries and overall cell morphology.

Mount coverslips using anti-fade mounting medium to preserve fluorescence during microscopy.

Microscopy and Image Analysis

Examine samples using a fluorescence microscope with appropriate filter sets for your fluorophores.

For each protein, assess:

- **Subcellular localization:** Nuclear vs. cytoplasmic
- **Staining intensity:** Relative levels between cell lines
- **Staining pattern:** Diffuse vs. punctate vs. excluded from certain regions

Take representative images at consistent exposure settings to allow qualitative comparisons between cell lines.

Part 6: Data Integration and Analysis

Now let's bring together all our data to understand the molecular profiles of these cell lines.

Protein Quantification and Normalization

For western blotting data, quantify band intensities using densitometry software. Normalize each protein of interest to the loading control in the same lane to account for loading differences.

Express results as relative protein levels compared to a reference sample or as ratios between different proteins.

Correlation Analysis

Look for correlations between:

- KRAS mutation status and protein levels
- Protein levels between different tumor suppressors
- Western blotting results and immunofluorescence patterns

These correlations can reveal functional relationships between different molecular alterations.

Expected Results and Biological Interpretation

Based on the known characteristics of these cell lines, you should expect:

HeLa cells:

- Wild-type KRAS
- Very low p53 levels (due to HPV E6-mediated degradation)
- Normal APC levels
- Low RB levels (due to HPV E7 interaction)

MCF-7 cells:

- Wild-type KRAS
- Wild-type p53 levels (these cells have functional p53)
- Normal APC levels
- Normal RB levels

HCT116 cells:

- KRAS G13D mutation
- Wild-type p53 levels (functional p53 pathway)
- Truncated APC protein (may show altered localization)
- Normal RB levels

Connecting Molecular Data to Cellular Behavior

Consider how these molecular profiles relate to the cellular behaviors you observed last week:

- Do more aggressive cell lines have more tumor suppressor losses?
- Do KRAS-mutant cells show different growth patterns?
- How do protein localization patterns relate to function?

Quality Control and Troubleshooting

Throughout today's experiments, several issues could affect your results:

PCR Troubleshooting

- **No bands:** Check DNA quality, primer concentrations, thermocycling parameters
- **Multiple bands:** Reduce annealing temperature or primer concentration
- **Weak bands:** Increase template DNA or cycle number

Western Blotting Issues

- **No signal:** Check antibody concentrations, transfer efficiency, detection reagents
- **High background:** Increase blocking time, reduce antibody concentrations
- **Uneven loading:** Improve protein quantification, check pipetting accuracy

Immunofluorescence Problems

- **No staining:** Check antibody activity, permeabilization, fixation conditions
- **High background:** Increase washing, reduce antibody concentrations
- **Poor morphology:** Optimize fixation time and conditions

Clinical and Research Relevance

The techniques you're learning today are directly applicable to both research and clinical settings:

Research Applications

- Characterizing new cell lines or tumor samples
- Studying the effects of therapeutic interventions
- Investigating molecular mechanisms of drug resistance
- Validating targets for drug development

Clinical Applications

- Mutation testing for treatment selection
- Biomarker analysis for prognosis
- Monitoring treatment response
- Identifying resistance mechanisms

Emerging Technologies

While the techniques we're using today remain standard in many laboratories, new technologies are expanding our analytical capabilities:

- Next-generation sequencing for comprehensive mutation analysis
- Mass spectrometry for quantitative proteomics
- High-content imaging for automated analysis
- Single-cell techniques for studying tumor heterogeneity

Preparation for Next Week

Save aliquots of your DNA and protein samples for future experiments. Store DNA at -20 degrees and proteins at -80 degrees with appropriate preservatives.

Document all your results carefully, including:

- Gel images with molecular weight markers
- Sequencing chromatograms with mutation calls
- Western blot images with quantification
- Immunofluorescence images with exposure settings

We'll use this data next week when we examine enabling mechanisms like genomic instability and inflammation.

Conclusion

Today's laboratory has given you hands-on experience with fundamental techniques in molecular cancer biology. You've learned to detect oncogene mutations, quantify tumor suppressor proteins, and analyze subcellular protein localization.

More importantly, you've begun to see how different molecular alterations contribute to the cancer phenotype. The integration of DNA and protein analysis provides a comprehensive view of cellular transformation that no single technique could provide alone.

The molecular profiles you've generated today will serve as a foundation for understanding more complex aspects of cancer biology as we progress through the course. Remember that these techniques represent the tools that have enabled our current understanding of cancer and continue to drive discoveries in cancer research.

Next week, we'll build on this molecular foundation to examine how genomic instability and inflammation enable the acquisition of cancer hallmarks. The groundwork you've laid today will help you understand how these enabling mechanisms interact with oncogene activation and tumor suppressor loss to drive cancer development.

Take detailed notes on your observations and results—they'll be valuable for your laboratory reports and for connecting this practical work to the theoretical concepts we discuss in lectures.

Excellent work today, and I look forward to seeing your results and discussing their implications.